THE EXAM WORKBOOK

Grade 9 General

افهم الفكرة. أتقن الحل.

Expressions. Functions. Sequences.

26 learning outcomes

104 focused questions

60 minute online mock

Name / الاسم _____ Class / الصف _____

A short route to confident practice

READ → REASON → CALCULATE → CHECK. Calculator guides on the website use fx-991EX ClassWiz.

اقرأ القاعدة، حلّ بنفسك، ثم راجع خطواتك. لكل مخرج 4 أسئلة مختارة، الرسومات من الكتاب الأصلي.

Your topic map

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Selected from the uploaded Reveal textbook, printed pp. 1–281, aligned to the supplied 26 Grade 9 General learning outcomes. Written tasks and original diagrams come from the book. MCQs include exam-style adaptations and labelled concept/example practice. This is a study selection, not an official exam or a prediction. The official coverage allows a scientific calculator and specifies 150 minutes / 100 marks. Online mock: 20 MCQs / 60 minutes. Calculator: fx-991EX ClassWiz.

معطيات المقالية من الكتاب، والاختيار من متعدد يتضمن تطبيقات موثقة للمفاهيم والأمثلة. لا يمكن ضمان ورود أسئلة بعينها في الاختبار النهائي.

01 / CHOOSE & CHECK

Q1 — Q2

Q1 Numerical expressions
التعابير العددية

Remember Group words such as sum and difference with parentheses.

1.1 MCQ

Which numerical expression represents the sum of 3 and 7, divided by 2?

Reveal · p. 9 · 11 · MCQ adaptation

A $\frac{3}{7+2}$ B $2(3+7)$
C $\frac{3+7}{2}$ D $3+\frac{7}{2}$

1.2 MCQ

Which numerical expression represents the difference of 6 and 2, divided by 4?

Reveal · p. 9 · 12 · MCQ adaptation

A $4(6-2)$ B $\frac{6-2}{4}$
C $6-\frac{2}{4}$ D $\frac{6+2}{4}$

1.3 MCQ

Which numerical expression represents the product of 3 and the sum of 4 and 9?

Reveal · p. 9 · 13 · MCQ adaptation

A $3(4+9)$ B $3(4)+9$
C $3+4+9$ D $4(3+9)$

1.4 MCQ

Which numerical expression represents 18 divided by the sum of 2 and 7?

Reveal · p. 9 · 14 · MCQ adaptation

A $\frac{18}{2}+7$ B $\frac{18+2}{7}$
C $18(2+7)$ D $\frac{18}{2+7}$

Q2 Distributive Property
خاصية التوزيع

Remember a(b+c)=ab+ac; distribute negative factors carefully.

2.1 MCQ

Use the Distributive Property to simplify.

$$2(x+4)$$

Reveal · p. 42 · 23 · MCQ adaptation

A $2x$ B $2x+8$
C $2x-8$ D 8

2.2 MCQ

Use the Distributive Property to simplify.

$$(5+n)3$$

Reveal · p. 42 · 24 · MCQ adaptation

A $3n+15$ B $3n$
C $3n-15$ D 15

2.3 MCQ

Use the Distributive Property to simplify.

$$(4-3m)8$$

Reveal · p. 42 · 25 · MCQ adaptation

A $-24m$ B $-24m-32$
C 32 D $32-24m$

2.4 MCQ

Use the Distributive Property to simplify.

$$-3(2x-6)$$

Reveal · p. 42 · 26 · MCQ adaptation

A $-6x$ B $-6x-18$
C $18-6x$ D 18

01 / CHOOSE & CHECK

Q3 — Q4

Q3 Addition and subtraction equations

معادلات الجمع والطرح

Remember Apply the same addition or subtraction to both sides.

3.1 MCQ

Which value satisfies the equation?

$$v - 9 = 14$$

Reveal · p. 81 · 1 · MCQ adaptation

A 23
B 5
C -23
D 14

3.2 MCQ

Which value satisfies the equation?

$$44 = t - 72$$

Reveal · p. 81 · 2 · MCQ adaptation

A -28
B 28
C -116
D 116

3.3 MCQ

Which value satisfies the equation?

$$-61 = d + (-18)$$

Reveal · p. 81 · 3 · MCQ adaptation

A -79
B 43
C -43
D 79

3.4 MCQ

Which value satisfies the equation?

$$18 + z = 40$$

Reveal · p. 81 · 4 · MCQ adaptation

A 58
B 22
C -22
D 40

Q4 Rearrange formulas

إعادة ترتيب الصيغ

Remember Isolate the requested variable; state nonzero divisor assumptions.

4.1 MCQ

Solve for v. Assume any divisor is nonzero. Which expression is correct?

$$u = vw + z$$

Reveal · p. 125 · 11 · MCQ adaptation

A $-\frac{z}{w}$
B $-\frac{u}{w} - \frac{z}{w}$
C $\frac{u}{w}$
D $\frac{u-z}{w}$

4.2 MCQ

Solve for c. Assume any divisor is nonzero. Which expression is correct?

$$x = b - cd$$

Reveal · p. 125 · 12 · MCQ adaptation

A $-\frac{x}{d}$
B $-\frac{b}{d} - \frac{x}{d}$
C $\frac{b-x}{d}$
D $\frac{b}{d}$

4.3 MCQ

Solve for g. Assume any divisor is nonzero. Which expression is correct?

$$fg - 9h = 10j$$

Reveal · p. 125 · 13 · MCQ adaptation

A $\frac{10j}{f}$
B $\frac{9h+10j}{f}$
C $-\frac{9h}{f} + \frac{10j}{f}$
D $\frac{9h}{f}$

4.4 MCQ

Solve for m. Assume any divisor is nonzero. Which expression is correct?

$$10m - p = -n$$

Reveal · p. 125 · 14 · MCQ adaptation

A $-\frac{n}{10} + \frac{p}{10}$
B $\frac{p}{10}$
C $\frac{n}{10} + \frac{p}{10}$
D $-\frac{n}{10}$

01 / CHOOSE & CHECK

Q5 — Q6

Q5 Properties of equality

خصائص المساواة

Remember Reflexive A=A; symmetric A=B⇒B=A; transitive A=B,B=C⇒A=C.

5.1 MCQ

Name the property illustrated.

$$4 + 17 = 21 \;\Rightarrow\; 21 = 4 + 17$$

Reveal · p. 31 · 1 · MCQ adaptation

- A Transitive Property of Equality
- B Additive Identity Property
- C Symmetric Property of Equality
- D Reflexive Property of Equality

5.2 MCQ

Name the property illustrated.

$$x + 3 = x + 3$$

Reveal · p. 31 · 2 · MCQ adaptation

- A Additive Identity Property
- B Reflexive Property of Equality
- C Symmetric Property of Equality
- D Transitive Property of Equality

5.3 MCQ

Name the property illustrated.

$$16 = 9 + 7 \;\Rightarrow\; 9 + 7 = 16$$

Reveal · p. 31 · 3 · MCQ adaptation

- A Symmetric Property of Equality
- B Reflexive Property of Equality
- C Transitive Property of Equality
- D Additive Identity Property

5.4 MCQ

Name the property illustrated.

$$6 + 2 = 4 + 4, \quad 4 + 4 = 8 \;\Rightarrow\; 6 + 2 = 8$$

Reveal · p. 31 · 4 · MCQ adaptation

- A Reflexive Property of Equality
- B Symmetric Property of Equality
- C Additive Identity Property
- D Transitive Property of Equality

Q6 Variables on each side

متغيرات في طرفي المعادلة

Remember Collect variable terms on one side and constants on the other.

6.1 MCQ

Which value satisfies the equation?

$$7c + 12 = -4c + 78$$

Reveal · p. 97 · 1 · MCQ adaptation

- A 30
- B 6
- C -6
- D $\frac{90}{11}$

6.2 MCQ

Which value satisfies the equation?

$$2m - 13 = -8m + 27$$

Reveal · p. 97 · 2 · MCQ adaptation

- A 4
- B $\frac{7}{5}$
- C $-\frac{20}{3}$
- D -4

6.3 MCQ

Which value satisfies the equation?

$$9x - 4 = 2x + 3$$

Reveal · p. 97 · 3 · MCQ adaptation

- A -1
- B $\frac{7}{11}$
- C $-\frac{1}{7}$
- D 1

6.4 MCQ

Which value satisfies the equation?

$$6 + 3t = 8t - 14$$

Reveal · p. 97 · 4 · MCQ adaptation

- A -4
- B $\frac{20}{11}$
- C 4
- D $-\frac{8}{5}$

01 / CHOOSE & CHECK

Q7

Q7 Distribute to evaluate

التوزيع للحساب الذهني

Remember Rewrite a factor as a nearby convenient value plus or minus a small number.

7.1 MCQ

Choose the correct distributive rewrite or estimate.

$5(99)$

Reveal · p. 37 · Example 2a · MCQ adaptation

A $5(100) - 5(1) = 495$
B $5(100) + 5(1) = 505$
C $5(100) - 1 = 499$
D $5(90) + 5(1) = 455$

7.2 MCQ

Choose the correct distributive rewrite or estimate.

$4(1002)$

Reveal · p. 37 · Example 2b · MCQ adaptation

A $4(1000) + 2 = 4002$
B $4(1000) - 4(2) = 3992$
C $4(100) + 4(2) = 408$
D $4(1000) + 4(2) = 4008$

7.5 MCQ

Which value is obtained by using the Distributive Property?

$6(99)$

Reveal · p. 37 · Application of Example 2a · Example variant

A 594
B 600
C 599
D 606

7.6 MCQ

Which value is obtained by using the Distributive Property?

$8(102)$

Reveal · p. 37 · Application of Example 2a · Example variant

A 808
B 800
C 816
D 802

01 / CHOOSE & CHECK

Q8

Q8 Rate of change
معدل التغير

Remember A linear function has a constant Δy/Δx.

8.1 MCQ

Is the function linear? State its rate of change.

Reveal · p. 221 · Example 3 · MCQ adaptation

x	y
11	−5
8	−3
5	−1
2	1
−1	3

A Linear; rate 2/3.
B Linear; rate −3/2.
C Nonlinear; rate is not constant.
D Linear; rate −2/3.

8.2 MCQ

Is the function linear?

Reveal · p. 221 · Example 4 · MCQ adaptation

x	y
22	−4
29	−1
36	1
43	4
50	6

A Yes; the rate is 2/7.
B Yes; the rate is 7/3.
C No; successive rates alternate between 3/7 and 2/7.
D Yes; the rate is 3/7.

8.5 MCQ

Which statement describes the rate of change in the listed data?

$(0, 1),\ (2, 7),\ (4, 13)$

Reveal · p. 221 · Application of Example 3 · Example variant

A 3
B -3
C $\frac{1}{3}$
D 6

8.6 MCQ

Which statement describes the rate of change in the listed data?

$(0, 9),\ (2, 5),\ (4, 1)$

Reveal · p. 221 · Application of Example 3 · Example variant

A -2
B 2
C $-\frac{1}{2}$
D -4

01 / CHOOSE & CHECK

Q9 — Q10

Q9 Solve proportions
حل التناسب

Remember For b,d≠0, a/b=c/d implies ad=bc.

9.1 MCQ

Which value satisfies the proportion?

$$\frac{x-3}{5} = \frac{6}{10}$$

Reveal · p. 113 · 13 · MCQ adaptation

A 0 B 3
C 6 D 15

9.2 MCQ

Which value satisfies the proportion?

$$\frac{7}{x+9} = \frac{21}{36}$$

Reveal · p. 113 · 14 · MCQ adaptation

A -3 B 3
C 12 D 21

9.3 MCQ

Which value satisfies the proportion?

$$\frac{10}{15} = \frac{4}{x-5}$$

Reveal · p. 113 · 15 · MCQ adaptation

A 11 B 1
C 6 D $\frac{23}{3}$

9.4 MCQ

Solve. Round to the nearest hundredth.

$$\frac{6}{14} = \frac{7}{x-3}$$

Reveal · p. 113 · 16 · MCQ adaptation

A 13.33 B 16.33
C 9.00 D 19.33

Q10 Multi-step equations with parameters
معادلات متعددة الخطوات بمعاملات

Remember Undo constants before dividing by a nonzero coefficient.

10.1 MCQ

Solve for x. Assume a≠0. Which expression is correct?

$$ax + 3 = 23$$

Reveal · p. 89 · 18 · MCQ adaptation

A $\frac{26}{a}$ B $\frac{20}{a}$
C $20a$ D 20

10.2 MCQ

Solve for x. Assume a≠0. Which expression is correct?

$$4 = ax - 14$$

Reveal · p. 89 · 19 · MCQ adaptation

A $\frac{18}{a}$ B $-\frac{10}{a}$
C $18a$ D 18

10.3 MCQ

Solve for x. Assume a≠0. Which expression is correct?

$$ax - 5 = 19$$

Reveal · p. 89 · 20 · MCQ adaptation

A $\frac{14}{a}$ B $24a$
C 24 D $\frac{24}{a}$

10.4 MCQ

Solve for x. Assume a≠0. Which expression is correct?

$$6 + ax = -29$$

Reveal · p. 89 · 21 · MCQ adaptation

A $-\frac{23}{a}$ B $-35a$
C $-\frac{35}{a}$ D $\frac{35}{a}$

01 / CHOOSE & CHECK

Q11

Q11 Dilations of linear functions
التمدد لدوال خطية

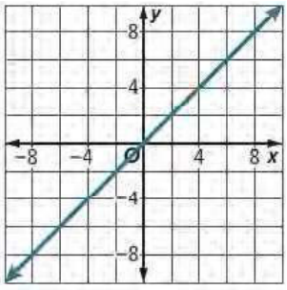
Remember Relative to y=x, kx stretches vertically if |k|>1 and compresses if 0<|k|<1.

11.1 MCQ

Describe the dilation relative to the parent function f(x)=x.

$$g(x) = 5x$$

Reveal · p. 248 · 10 · MCQ adaptation



A Vertical stretch by factor 5.

B Horizontal shift right by 5.

C Vertical shift up by 5.

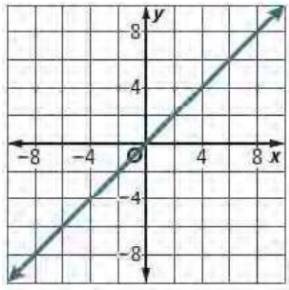
D Vertical compression by factor 5.

11.2 MCQ

Describe the dilation relative to the parent function f(x)=x.

$$g(x) = \frac{1}{3}x$$

Reveal · p. 248 · 11 · MCQ adaptation



A Horizontal shift right by 1/3.

B Vertical shift up by 1/3.

C Vertical stretch by factor 1/3.

D Vertical compression by factor 1/3.

11.5 MCQ

For g(x)=3x relative to f(x)=x, which statement is correct?

Reveal · p. 248 · Application of 10 · Concept application

A Vertical stretch by 3.

B Vertical compression by 3.

C Translation up 3.

D Translation right 3.

11.6 MCQ

For g(x)=0.4x relative to f(x)=x, which statement is correct?

Reveal · p. 248 · Application of 10 · Concept application

A Translation down 0.4.

B Vertical compression by 0.4.

C Vertical stretch by 0.4.

D Reflection only.

01 / CHOOSE & CHECK

Q12

Q12 Slope-intercept form
صيغة الميل والمقطع

Remember Isolate y to obtain y=mx+b.

12.1 MCQ

Which equation is equivalent to the given equation in slope-intercept form?

$$-10x + 2y = 12$$

Reveal · p. 235 · 9 · MCQ adaptation

A $y = 5x$
B $y = 5x - 6$
C $y = 6$
D $y = 5x + 6$

12.2 MCQ

Which equation is equivalent to the given equation in slope-intercept form?

$$4y + 12x = 16$$

Reveal · p. 235 · 10 · MCQ adaptation

A $y = -3x$
B $y = -3x - 4$
C $y = 4 - 3x$
D $y = 4$

12.3 MCQ

Which equation is equivalent to the given equation in slope-intercept form?

$$-5x + 15y = -30$$

Reveal · p. 235 · 11 · MCQ adaptation

A $y = \frac{x}{3}$
B $y = \frac{x}{3} - 2$
C $y = \frac{x}{3} + 2$
D $y = -2$

12.4 MCQ

Which equation is equivalent to the given equation in slope-intercept form?

$$6x - 3y = -18$$

Reveal · p. 235 · 12 · MCQ adaptation

A $y = 2x + 6$
B $y = 2x$
C $y = 2x - 6$
D $y = 6$

01 / CHOOSE & CHECK

Q13

Q13 Read linear graphs

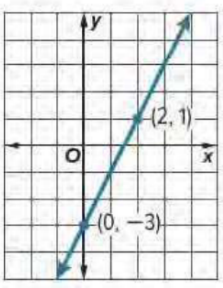
قراءة الرسوم الخطية

Remember Slope is rise/run; b is where the line crosses the y-axis.

13.1 MCQ

Which equation matches the graph?

Reveal · p. 236 · 37 · MCQ adaptation



A $y = 2x$

B $y = 2x + 3$

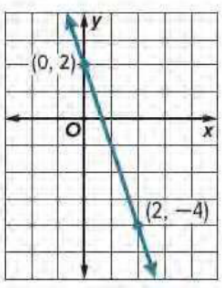
C $y = 2x - 3$

D $y = -3$

13.2 MCQ

Which equation matches the graph?

Reveal · p. 236 · 38 · MCQ adaptation



A $y = -3x$

B $y = 2 - 3x$

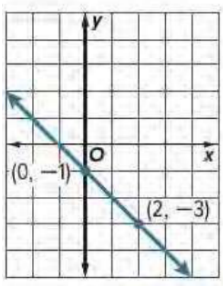
C $y = -3x - 2$

D $y = 2$

13.3 MCQ

Which equation matches the graph?

Reveal · p. 236 · 39 · MCQ adaptation



A $y = -x - 1$

B $y = -x$

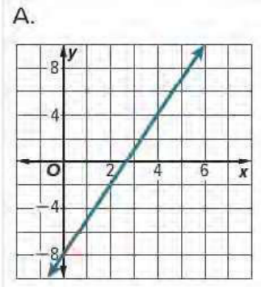
C $y = 1 - x$

D $y = -1$

13.4 MCQ

Match graph A with its equation.

Reveal · p. 233 · Match graph A · MCQ adaptation



A $y = 3x$

B $y = 3x + 8$

C $y = -8$

D $y = 3x - 8$

01 / CHOOSE & CHECK

Q14

Q14 Graph step functions
رسم الدوال الدرجية

Remember ⌊x⌋ is the greatest integer ≤x; pay attention to open and closed endpoints.

14.1 MCQ

Choose the correct graph description or domain and range.

$$f(x) = 3\lfloor x \rfloor$$

Reveal · p. 263 · 7 · MCQ adaptation

A $y = n$ on $[3n, 3n+3)$
B $y = 3n$ on $[n, n+1),\ n \in \mathbb{Z}$
C $y = 3n$ on $(n, n+1]$
D $y = 3x$

14.2 MCQ

Choose the correct graph description or domain and range.

$$f(x) = \lfloor -x \rfloor$$

Reveal · p. 263 · 8 · MCQ adaptation

A $y = n$ on $(-n-1, -n],\ n \in \mathbb{Z}$
B $y = n$ on $[-n-1, -n)$
C $y = -x$
D $y = n$ on $[n, n+1)$

14.5 MCQ

For f(x)=floor(x), which point belongs to the graph?

Reveal · p. 263 · Application of 7 · Concept application

A (0, 1)
B (−1.2, −2)
C (−1.2, −1)
D (1.2, 2)

14.6 MCQ

For f(x)=floor(x), how are endpoints drawn on the step y=2?

Reveal · p. 263 · Application of 7 · Concept application

A Closed at x=2; open at x=3.
B Open at x=2; closed at x=3.
C Closed at both endpoints.
D Open at both endpoints.

01 / CHOOSE & CHECK

Q15

Q15 Slope from two points
الميل بين نقطتين

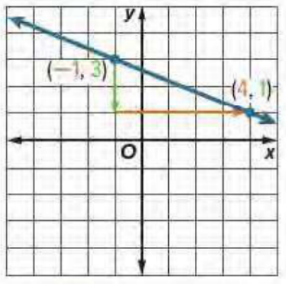
Remember $m=(y_2-y_1)/(x_2-x_1)$; vertical lines have undefined slope.

15.1 MCQ

What is the slope of the line through the given points?

$(-1,3),\ (4,1)$

Reveal · p. 223 · Example 6 · MCQ adaptation



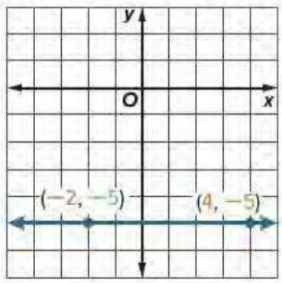
A $-\frac{2}{5}$
B $\frac{2}{5}$
C $-\frac{5}{2}$
D $-\frac{2}{3}$

15.2 MCQ

What is the slope of the line through the given points?

$(-2,-5),\ (4,-5)$

Reveal · p. 223 · Example 7 · MCQ adaptation



A 5
B 1
C Undefined
D 0

15.5 MCQ

What is the slope of the line through the two points?

$(1,2),\ (3,6)$

Reveal · p. 223 · Application of Example 6 · Example variant

A 2
B -2
C $\frac{1}{2}$
D 4

15.6 MCQ

What is the slope of the line through the two points?

$(1,5),\ (3,-1)$

Reveal · p. 223 · Application of Example 6 · Example variant

A 3
B $-\frac{1}{3}$
C -6
D -3

01 / CHOOSE & CHECK

Q16

Q16 Evaluate step functions

إيجاد قيم الدوال الدرجية

Remember Evaluate the expression inside the floor brackets first.

16.1 MCQ

What is the value of the given expression? Evaluate the step function at the given value.

$$f(x) = 2\lfloor x - 1 \rfloor, \quad x = 1.5$$

Reveal · p. 265 · 18a · MCQ adaptation

A 1
B 2
C -2
D 0

16.2 MCQ

What is the value of the given expression? Evaluate the step function at the given value.

$$f(x) = 2\lfloor x - 1 \rfloor, \quad x = 2.2$$

Reveal · p. 265 · 18b · MCQ adaptation

A 2.4
B 4
C 2
D 1

16.3 MCQ

What is the value of the given expression? Evaluate the step function at the given value.

$$f(x) = 2\lfloor x - 1 \rfloor, \quad x = 9.7$$

Reveal · p. 265 · 18c · MCQ adaptation

A 17.4
B 16
C 18
D 8

16.4 MCQ

What is the value of the given expression? Evaluate the step function at the given value.

$$f(x) = 2\lfloor x - 1 \rfloor, \quad x = -1.25$$

Reveal · p. 265 · 18d · MCQ adaptation

A -6
B -4
C -4.5
D 6

01 / CHOOSE & CHECK

Q17

Q17 Arithmetic sequence arguments

تبرير المتتاليات الحسابية

Remember Check that all consecutive differences are equal.

17.1 MCQ

Determine whether the sequence is arithmetic. Choose the justification.

$$-9, -12, -15, -18, \ldots$$

Reveal · p. 257 · 28 · MCQ adaptation

A Not arithmetic: increasing terms cannot be arithmetic.

B Arithmetic: the common difference is 1.

C Arithmetic: every difference is −3.

D Arithmetic: a constant ratio is sufficient.

17.2 MCQ

Determine whether the sequence is arithmetic. Choose the justification.

$$10, 15, 25, 40, \ldots$$

Reveal · p. 257 · 29 · MCQ adaptation

A Arithmetic: the common difference is 1.

B Not arithmetic: differences 5, 10, 15 are unequal.

C Arithmetic: a constant ratio is sufficient.

D Not arithmetic: increasing terms cannot be arithmetic.

17.5 MCQ

What are the next three terms?

$$-9, -12, -15, -18, \ldots$$

Reveal · p. 257 · Application of 28 · Example variant

A $-20, -22, -24$

B $-15, -12, -9$

C $-19, -20, -21$

D $-21, -24, -27$

17.6 MCQ

Which formula gives the nth term?

$$-9, -12, -15, -18, \ldots$$

Reveal · p. 257 · Application of 28 · Example variant

A $a_n = -3n - 6$

B $a_n = -3n - 9$

C $a_n = 3n - 12$

D $a_n = -6n - 3$

01 / CHOOSE & CHECK

Q18

Q18 Constant linear functions
الدوال الخطية الثابتة

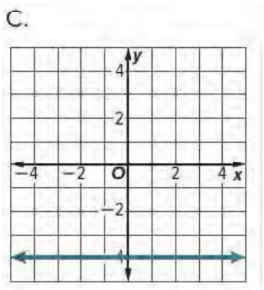
Remember y=b is horizontal, with slope 0 and y-intercept b.

18.1 MCQ

Match the original constant-function graph with its equation.

Reveal · p. 233 · Match graph C · MCQ adaptation

C.



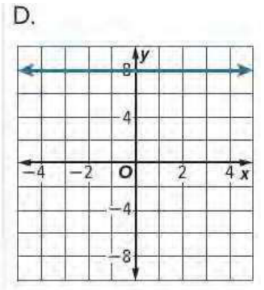
A $y = -4x$
B $y = -4$
C $y = -3$
D $y = 4$

18.2 MCQ

Match the original constant-function graph with its equation.

Reveal · p. 233 · Match graph D · MCQ adaptation

D.



A $y = 8$
B $y = 9$
C $y = -8$
D $y = 8x$

18.5 MCQ

Which equation has a horizontal graph through (0,−3)?

Reveal · p. 233 · Application of Match graph C · Concept application

A y=−3x
B y=x−3
C y=−3
D x=−3

18.6 MCQ

What is the slope of y=8?

Reveal · p. 233 · Application of Match graph C · Concept application

A 1
B Undefined
C 0
D 8

01 / CHOOSE & CHECK

Q19

Q19 Verbal expressions
التعابير اللفظية

Remember Translate each operation and preserve subtraction/division order.

19.1 MCQ

Which verbal expression represents the given algebraic expression?

$4q$

Reveal · p. 18 · 1 · MCQ adaptation

A The product of 4 and q.
B The sum of 4 and q.
C q divided by 4.
D 4 less than q.

19.2 MCQ

Which verbal expression represents the given algebraic expression?

$\frac{1}{8}y$

Reveal · p. 18 · 2 · MCQ adaptation

A Eight times y.
B y minus eight.
C Eight divided by y.
D One eighth of y.

19.3 MCQ

Which verbal expression represents the given algebraic expression?

$15 + r$

Reveal · p. 18 · 3 · MCQ adaptation

A 15 minus r.
B r divided by 15.
C The sum of 15 and r.
D The product of 15 and r.

19.4 MCQ

Which verbal expression represents the given algebraic expression?

$w - 24$

Reveal · p. 18 · 4 · MCQ adaptation

A w plus 24.
B 24 less than w.
C w less than 24.
D 24 times w.

01 / CHOOSE & CHECK

Q20

Q20

Solve multi-step equations

حل المعادلات متعددة الخطوات

Remember Distribute if needed, combine like terms, then use inverse operations.

20.1 MCQ

Which value satisfies the equation?

$$3x + 8 = 29$$

Reveal · p. 89 · 27 · MCQ adaptation

A $\frac{37}{3}$
B 21
C -7
D 7

20.2 MCQ

Which value satisfies the equation?

$$\frac{5f}{2} - 6 = 19$$

Reveal · p. 89 · 29 · MCQ adaptation

A $\frac{26}{5}$
B 25
C 10
D -10

20.3 MCQ

Which value satisfies the equation?

$$5(1 + n) = -5$$

Reveal · p. 89 · 33 · MCQ adaptation

A 0
B -2
C -1
D -6

20.4 MCQ

Which value satisfies the equation?

$$-27 = -6 - 3p$$

Reveal · p. 89 · 34 · MCQ adaptation

A 7
B -7
C 11
D 21

02 / WRITE THE STEPS

Q21 · Written practice 1

Q21

Evaluate algebraic expressions

إيجاد قيم التعابير الجبرية

Remember Substitute first, then powers, multiplication/division, addition/subtraction.

21.1 FRQ · show your working

Evaluate if g=2, r=3, and t=11.

$$g + 6t$$

Reveal · p. 19 · 42

(a) Show substitutions and evaluate.

21.2 FRQ · show your working

Evaluate if g=2, r=3, and t=11.

$$7 - gr$$

Reveal · p. 19 · 43

(a) Show substitutions and evaluate.

21.3 FRQ · show your working

Evaluate if g=2, r=3, and t=11.

$$r^2 + (g^3 - 8)^5$$

Reveal · p. 19 · 44

(a) Show substitutions and evaluate.

21.4 FRQ · show your working

Evaluate if g=2, r=3, and t=11.

$$(2t + 3g) \div 4$$

Reveal · p. 19 · 45

(a) Show substitutions and evaluate.

02 / WRITE THE STEPS

Q22 · Written practice 1

Q22

Use and rearrange formulas

تطبيق الصيغ وإعادة ترتيبها

Remember Rearrange symbolically before substituting numerical values.

22.1 FRQ · show your working

The perimeter of a rectangle is P=2ℓ+2w.

$$P = 2\ell + 2w$$

Reveal · p. 125 · 19

(a) Solve for ℓ.

(b) Find ℓ when w=4 m and P=36 m.

22.2 FRQ · show your working

A batting average is a=h/b, with h hits in b times at bat.

$$a = h/b$$

Reveal · p. 125 · 20

(a) Solve for b.

(b) Find b when a=0.325 and h=39.

02 / WRITE THE STEPS

Q22 · Written practice 2

Q22

Use and rearrange formulas

تطبيق الصيغ وإعادة ترتيبها

Remember Rearrange symbolically before substituting numerical values.

22.3 FRQ · show your working

Videogames cost $13.50 each. Controllers cost p dollars; c is the total cost.

$$c = 13.50g + p$$

Reveal · p. 125 · 21

(a) Solve for g.

(b) Find g when p=$55 and c=$136.

22.4 FRQ · show your working

The volume of a box is V=ℓwh.

$$V = \ell wh$$

Reveal · p. 125 · 22

(a) Solve for h.

(b) Find h when V=50 m³, ℓ=10 m, and w=2 m.

02 / WRITE THE STEPS

Q23 · Written practice 1

Q23

Equations, identities and contradictions

المعادلات والمتطابقات والتناقض

Remember A true constant statement is an identity; a false one means no solution.

23.1 FRQ · show your working

Solve the equation. Check your solution.

$$2(r+6)=4(r+4)$$

Reveal · p. 97 · 7

(a) Show your steps. State if there is one solution, no solution, or an identity.

23.2 FRQ · show your working

Solve the equation. Check your solution.

$$3(3m-2)=2(3m+3)$$

Reveal · p. 97 · 11

(a) Show your steps. State if there is one solution, no solution, or an identity.

23.3 FRQ · show your working

Solve the equation. Check your solution.

$$2x=2(x-3)$$

Reveal · p. 99 · 37

(a) Show your steps. State if there is one solution, no solution, or an identity.

23.4 FRQ · show your working

Solve the equation. Check your solution.

$$-5(3-q)+4=5q-11$$

Reveal · p. 99 · 39

(a) Show your steps. State if there is one solution, no solution, or an identity.

02 / WRITE THE STEPS

Q24 · Written practice 1

Q24 Graph with a table
الرسم باستخدام جدول

Remember Choose convenient x-values, calculate y, and plot collinear points.

24.1 FRQ · show your working

Graph the equation by making a table.

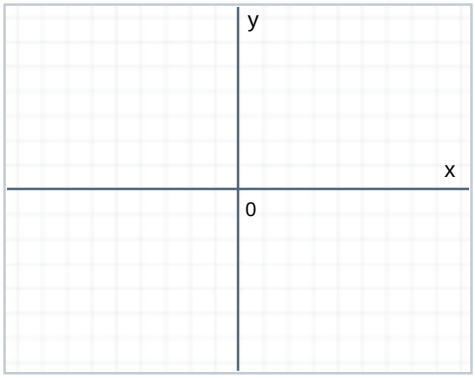
$$y - 8 = -x$$

Reveal · p. 215 · 5

(a) Make a table of values, then plot the points and draw the complete line.

fx-991EX · TABLE MENU → Table (9). SETUP → Table → f(x) only. Enter your equation using ALPHA X, then =. Set Start, End, Step; copy 3 pairs. Isolate y first.

x			
y			



24.2 FRQ · show your working

Graph the equation by making a table.

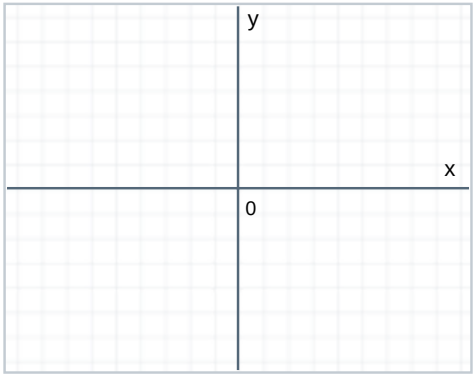
$$x = 10 - y$$

Reveal · p. 215 · 6

(a) Make a table of values, then plot the points and draw the complete line.

fx-991EX · TABLE MENU → Table (9). SETUP → Table → f(x) only. Enter your equation using ALPHA X, then =. Set Start, End, Step; copy 3 pairs. Isolate y first.

x			
y			



02 / WRITE THE STEPS

Q24 · Written practice 2

Q24

Graph with a table

الرسم باستخدام جدول

Remember Choose convenient x-values, calculate y, and plot collinear points.

24.3 FRQ · show your working

Graph the equation by making a table.

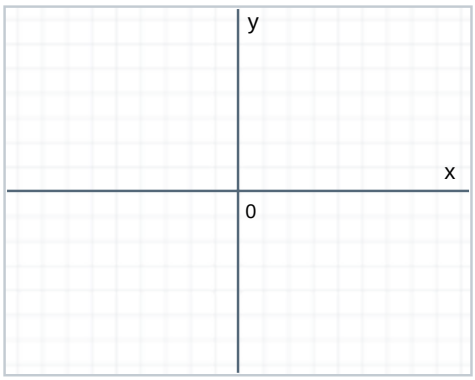
$$y = \frac{1}{2}x + 1$$

Reveal · p. 215 · 7

(a) Make a table of values, then plot the points and draw the complete line.

fx-991EX · TABLE MENU → Table (9). SETUP → Table → f(x) only. Enter your equation using ALPHA X, then =. Set Start, End, Step; copy 3 pairs. Isolate y first.

x			
y			



24.4 FRQ · show your working

Graph the equation by making a table.

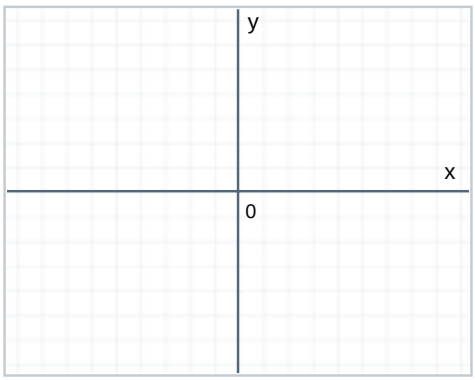
$$y + 2 = \frac{1}{4}x$$

Reveal · p. 215 · 8

(a) Make a table of values, then plot the points and draw the complete line.

fx-991EX · TABLE MENU → Table (9). SETUP → Table → f(x) only. Enter your equation using ALPHA X, then =. Set Start, End, Step; copy 3 pairs. Isolate y first.

x			
y			



02 / WRITE THE STEPS

Q25 · Written practice 1

Q25

Piecewise functions

الدوال متعددة التعريف

Remember Use each formula only on its stated interval; show endpoint inclusion.

25.1 FRQ · show your working

Graph the function. State its domain and range.

$$f(x) = \begin{cases} \frac{1}{2}x - 1 & x > 3 \\ -2x + 3 & x \leq 3 \end{cases}$$

Reveal · p. 263 · 1

(a) Plot both pieces with correct endpoints; state domain and range.

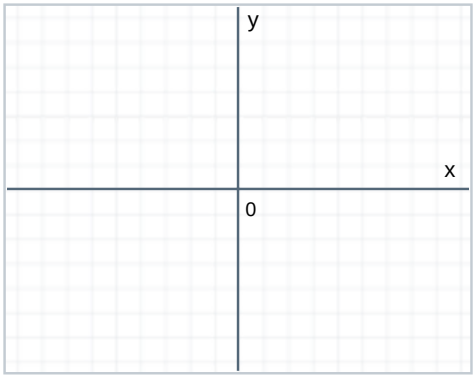
fx-991EX · TABLE MENU → Table (9). SETUP → Table → f(x) only. Enter your equation using ALPHA X, then =. Set Start, End, Step; copy 3 pairs. Isolate y first. Repeat for each branch; respect its domain and open/closed endpoint.

Branch 1

x			
y			

Branch 2

x			
y			



25.2 FRQ · show your working

Graph the function. State its domain and range.

$$f(x) = \begin{cases} 2x - 5 & x > 1 \\ 4x - 3 & x \leq 1 \end{cases}$$

Reveal · p. 263 · 2

(a) Plot both pieces with correct endpoints; state domain and range.

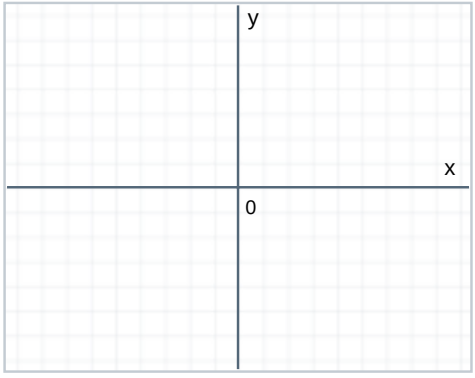
fx-991EX · TABLE MENU → Table (9). SETUP → Table → f(x) only. Enter your equation using ALPHA X, then =. Set Start, End, Step; copy 3 pairs. Isolate y first. Repeat for each branch; respect its domain and open/closed endpoint.

Branch 1

x			
y			

Branch 2

x			
y			



02 / WRITE THE STEPS

Q25 · Written practice 2

Q25 Piecewise functions
الدوال متعددة التعريف

Remember Use each formula only on its stated interval; show endpoint inclusion.

25.3 FRQ · show your working

Graph the function. State its domain and range.

$$f(x) = \begin{cases} 2x + 3 & x \geq -3 \\ \frac{1}{3}x + 1 & x < -3 \end{cases}$$

Reveal · p. 263 · 3

(a) Plot both pieces with correct endpoints; state domain and range.

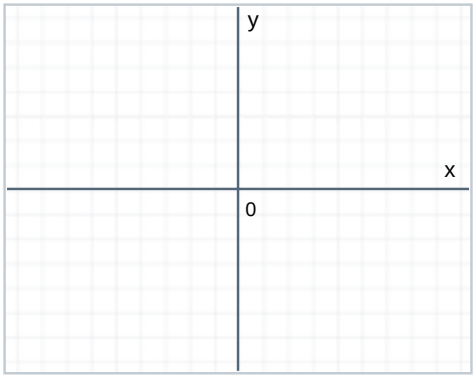
fx-991EX · TABLE MENU → Table (9). SETUP → Table → f(x) only. Enter your equation using ALPHA X, then =. Set Start, End, Step; copy 3 pairs. Isolate y first. Repeat for each branch; respect its domain and open/closed endpoint.

Branch 1

x			
y			

Branch 2

x			
y			



25.4 FRQ · show your working

Graph the function. State its domain and range.

$$f(x) = \begin{cases} 3x + 4 & x \geq 1 \\ x + 3 & x < 1 \end{cases}$$

Reveal · p. 263 · 4

(a) Plot both pieces with correct endpoints; state domain and range.

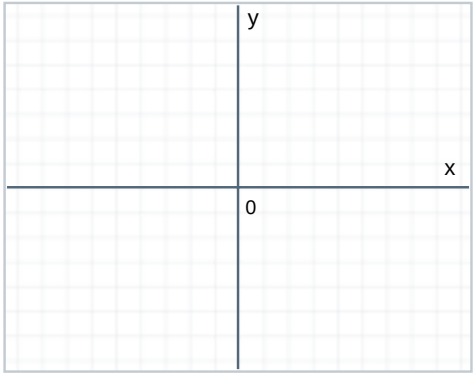
fx-991EX · TABLE MENU → Table (9). SETUP → Table → f(x) only. Enter your equation using ALPHA X, then =. Set Start, End, Step; copy 3 pairs. Isolate y first. Repeat for each branch; respect its domain and open/closed endpoint.

Branch 1

x			
y			

Branch 2

x			
y			



02 / WRITE THE STEPS

Q26 · Written practice 1

Q26

Arithmetic nth term

الحد النوني للمتتالية الحسابية

Remember $a_n=a_1+(n-1)d$; substitute the requested term number.

26.1 FRQ · show your working

Use the arithmetic sequence to write an equation and find the 7th term.

$$-3, -8, -13, -18, \ldots$$

Reveal · p. 255 · 19

(a) Write a formula for the nth term.

(b) Find the 7th term.

26.2 FRQ · show your working

Use the arithmetic sequence to write an equation and find the 7th term.

$$-2, 3, 8, 13, \ldots$$

Reveal · p. 255 · 20

(a) Write a formula for the nth term.

(b) Find the 7th term.

02 / WRITE THE STEPS

Q26 · Written practice 2

Q26

Arithmetic nth term

الحد النوني للمتتالية الحسابية

Remember $a_n=a_1+(n-1)d$; substitute the requested term number.

26.3 FRQ · show your working

Use the arithmetic sequence to write an equation and find the 7th term.

$$-11, -15, -19, -23, \ldots$$

Reveal · p. 255 · 21

(a) Write a formula for the nth term.

(b) Find the 7th term.

26.4 FRQ · show your working

Use the arithmetic sequence to write an equation and find the 7th term.

$$-0.75, -0.5, -0.25, 0, \ldots$$

Reveal · p. 255 · 22

(a) Write a formula for the nth term.

(b) Find the 7th term.

03 / CHECK & LEARN

MCQ answer key

Check after solving. Match the printed question number to its correct option. Full explanations are available in the online LO bank.

Q1 · Numerical expressions
1.1: C 1.2: B 1.3: A 1.4: D

Q2 · Distributive Property
2.1: B 2.2: A 2.3: D 2.4: C

Q3 · Addition and subtraction equations
3.1: A 3.2: D 3.3: C 3.4: B

Q4 · Rearrange formulas
4.1: D 4.2: C 4.3: B 4.4: A

Q5 · Properties of equality
5.1: C 5.2: B 5.3: A 5.4: D

Q6 · Variables on each side
6.1: B 6.2: A 6.3: D 6.4: C

Q7 · Distribute to evaluate
7.1: A 7.2: D 7.5: A 7.6: C

Q8 · Rate of change
8.1: D 8.2: C 8.5: A 8.6: A

Q9 · Solve proportions
9.1: C 9.2: B 9.3: A 9.4: D

Q10 · Multi-step equations with parameters
10.1: B 10.2: A 10.3: D 10.4: C

Q11 · Dilations of linear functions
11.1: A 11.2: D 11.5: A 11.6: B

Q12 · Slope-intercept form
12.1: D 12.2: C 12.3: B 12.4: A

Q13 · Read linear graphs
13.1: C 13.2: B 13.3: A 13.4: D

Q14 · Graph step functions
14.1: B 14.2: A 14.5: B 14.6: A

Q15 · Slope from two points
15.1: A 15.2: D 15.5: A 15.6: D

Q16 · Evaluate step functions
16.1: D 16.2: C 16.3: B 16.4: A

Q17 · Arithmetic sequence arguments
17.1: C 17.2: B 17.5: D 17.6: A

Q18 · Constant linear functions
18.1: B 18.2: A 18.5: C 18.6: C

Q19 · Verbal expressions
19.1: A 19.2: D 19.3: C 19.4: B

Q20 · Solve multi-step equations
20.1: D 20.2: C 20.3: B 20.4: A

03 / CHECK & LEARN

Worked solutions 1

21.1 p. 19 · 42

(a) 2+6(11)=2+66=68. Substitution (3), order of operations (4), final answer (3).

68

21.2 p. 19 · 43

(a) 7−2(3)=7−6=1. Substitution (3), order of operations (4), final answer (3).

1

21.3 p. 19 · 44

(a) $3^2+(2^3-8)^5=9+0=9$. Substitution (3), order of operations (4), final answer (3).

9

21.4 p. 19 · 45

(a) (2·11+3·2)/4=(22+6)/4=7. Substitution (3), order of operations (4), final answer (3).

7

22.1 p. 125 · 19

(a) Correct isolation and formula (5).

$\ell = (P - 2w)/2$

(b) ℓ=(36−8)/2=14 m. Substitution (2), calculation and units (3).

22.2 p. 125 · 20

(a) Correct isolation and formula (5).

$b = h/a$

(b) b=39/0.325=120 times at bat. Substitution (2), calculation and units (3).

22.3 p. 125 · 21

(a) Correct isolation and formula (5).

$g = (c - p)/13.50$

(b) g=(136−55)/13.50=6 games. Substitution (2), calculation and units (3).

22.4 p. 125 · 22

(a) Correct isolation and formula (5).

$h = V/(\ell w)$

(b) h=50/(10·2)=2.5 m. Substitution (2), calculation and units (3).

23.1 p. 97 · 7

(a) 2r+12=4r+16 ⇒ −4=2r ⇒ r=−2. Both sides equal 8. Distribution (3), collection/simplification (3), conclusion and check (4).

$r = -2$

23.2 p. 97 · 11

(a) 9m−6=6m+6 ⇒ 3m=12 ⇒ m=4. Both sides equal 30. Distribution (3), collection/simplification (3), conclusion and check (4).

$m = 4$

23.3 p. 99 · 37

(a) 2x=2x−6 ⇒ 0=−6, a contradiction. No solution. Distribution (3), collection/simplification (3), conclusion and check (4).

$\varnothing$

23.4 p. 99 · 39

(a) −15+5q+4=5q−11 ⇒ 5q−11=5q−11. Identity: every real q is a solution. Distribution (3), collection/simplification (3), conclusion and check (4).

$q \in \mathbb{R}$

03 / CHECK & LEARN

Worked solutions 2

24.1 p. 215 · 5

(a) One possible table: (0,8), (4,4), (8,0). Valid table (4), accurately plotted points (3), straight line with arrows and labelled axes (3).

TABLE: (0, 8) · (4, 4) · (8, 0)

24.2 p. 215 · 6

(a) One possible table: (0,10), (5,5), (10,0). Valid table (4), accurately plotted points (3), straight line with arrows and labelled axes (3).

TABLE: (0, 10) · (5, 5) · (10, 0)

24.3 p. 215 · 7

(a) One possible table: (−2,0), (0,1), (2,2). Valid table (4), accurately plotted points (3), straight line with arrows and labelled axes (3).

TABLE: (-2, 0) · (0, 1) · (2, 2)

24.4 p. 215 · 8

(a) One possible table: (−4,−3), (0,−2), (4,−1). Valid table (4), accurately plotted points (3), straight line with arrows and labelled axes (3).

TABLE: (-4, -3) · (0, -2) · (4, -1)

25.1 p. 263 · 1

(a) Open (3,0.5), slope 1/2 to the right; closed (3,−3), slope −2 to the left. D=ℝ, R=[−3,∞). Slopes and rays (4), endpoints (2), domain (2), range (2).

TABLE: (3, 0.5) · (4, 1) · (5, 1.5) [x>3; boundary open]
(1, 1) · (2, -1) · (3, -3) [x≤3; boundary closed]

25.2 p. 263 · 2

(a) Open (1,−3), slope 2 to the right; closed (1,1), slope 4 to the left. D=ℝ, R=ℝ. Slopes and rays (4), endpoints (2), domain (2), range (2).

TABLE: (1, -3) · (2, -1) · (3, 1) [x>1; boundary open]
(-1, -7) · (0, -3) · (1, 1) [x≤1; boundary closed]

25.3 p. 263 · 3

(a) Closed (−3,−3), slope 2 to the right; open (−3,0), slope 1/3 to the left. D=ℝ, R=ℝ. Slopes and rays (4), endpoints (2), domain (2), range (2).

TABLE: (-3, -3) · (-2, -1) · (-1, 1) [x≥−3; boundary closed]
(-9, -2) · (-6, -1) · (-3, 0) [x<−3; boundary open]

25.4 p. 263 · 4

(a) Closed (1,7), slope 3 to the right; open (1,4), slope 1 to the left. D=ℝ, R=(−∞,4)∪[7,∞). Slopes and rays (4), endpoints (2), domain (2), range (2).

TABLE: (1, 7) · (2, 10) · (3, 13) [x≥1; boundary closed]
(-1, 2) · (0, 3) · (1, 4) [x<1; boundary open]

26.1 p. 255 · 19

(a) d=−5; a_n=−3+(n−1)(−5)=2−5n. Common difference (2), formula (3).

$$a_n = 2 - 5n$$

(b) Substitute n=7: a_7=-33. Substitution (2), correct value (3).

26.2 p. 255 · 20

(a) d=5; a_n=−2+5(n−1)=5n−7. Common difference (2), formula (3).

$$a_n = 5n - 7$$

(b) Substitute n=7: a_7=28. Substitution (2), correct value (3).

26.3 p. 255 · 21

(a) d=−4; a_n=−11−4(n−1)=−4n−7. Common difference (2), formula (3).

$$a_n = -4n - 7$$

(b) Substitute n=7: a_7=-35. Substitution (2), correct value (3).

26.4 p. 255 · 22

(a) d=0.25; a_n=−0.75+0.25(n−1)=0.25n−1. Common difference (2), formula (3).

$$a_n = \frac{n}{4} - 1$$

(b) Substitute n=7: a_7=3/4. Substitution (2), correct value (3).